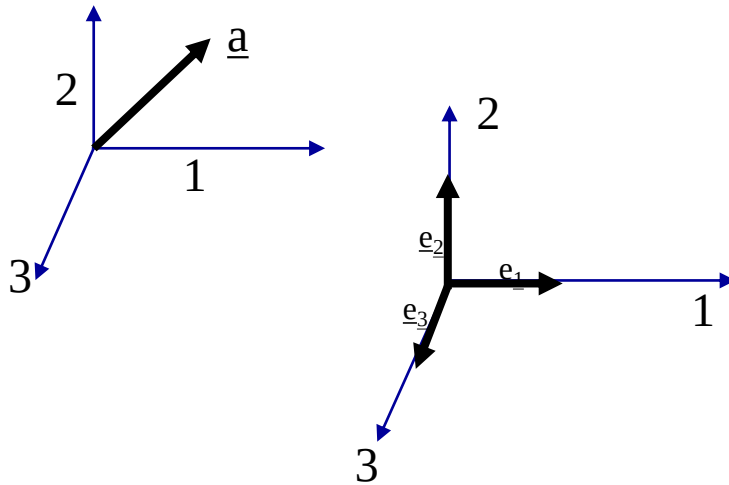


Lecture 3

ChE 467/596-012 Polymer Rheology

Saad A Khan

Last Class: Vectors and Tensors



Index Notation

$$\underline{a} = a_1 \underline{e}_1 + a_2 \underline{e}_2 + a_3 \underline{e}_3$$

$$\underline{a} = a_i \underline{e}_i$$

Kronecker delta

$$\delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$$\underline{e}_i \cdot \underline{e}_j = \delta_{ij}$$

Dot Product

$$\underline{V} \cdot \underline{W} = V_i \underline{e}_i \cdot W_j \underline{e}_j$$

$$= V_i W_j (\underline{e}_i \cdot \underline{e}_j)$$

$$= V_i W_j \delta_{ij}$$

$$= V_i W_i = V_1 W_1 + V_2 W_2 + V_3 W_3$$

Tensors

Scalar $a = a$

Vector $\underline{V} = V_i \underline{e}_i$

Tensor $\underline{\underline{T}} = T_{ij} \underline{e}_i \underline{e}_j \quad (\text{Dyad})$

$$\begin{aligned} &= T_{11} \underline{e}_1 \underline{e}_1 + T_{12} \underline{e}_1 \underline{e}_2 + T_{13} \underline{e}_1 \underline{e}_3 \\ &\quad + T_{21} \underline{e}_2 \underline{e}_1 + T_{22} \underline{e}_2 \underline{e}_2 + T_{23} \underline{e}_2 \underline{e}_3 \\ &\quad + T_{31} \underline{e}_3 \underline{e}_1 + T_{32} \underline{e}_3 \underline{e}_2 + T_{33} \underline{e}_3 \underline{e}_3 \end{aligned}$$

$$\underline{\underline{T}} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix}$$

Food for Thought

Two vectors: $\underline{a}$ and $\underline{b}$

Your thoughts on:

$$\underline{a} \cdot \underline{b}$$

$$\underline{a} \underline{b}$$

$$\underline{a} \times \underline{b}$$

Vectors & Tensors (last class)

$$\underline{t}_{(n)} = \underline{n} \cdot \underline{\underline{T}}$$

Do not use same index

$$\underline{n} \cdot \underline{\underline{T}} = n_i \underline{e}_i \cdot T_{jk} \underline{e}_j \underline{e}_k \quad \text{Take dot product of nearest vectors}$$

$$\underline{n} \cdot \underline{\underline{T}} = n_i \underline{e}_i \cdot T_{jk} \underline{e}_j \underline{e}_k$$

Nesting convention

$$= n_i \delta_{ij} T_{jk} \underline{e}_k = n_i T_{ik} \underline{e}_k \text{ or } n_j T_{jk} \underline{e}_k \text{ (repeated indices)}$$

$$= n_i T_{i1} \underline{e}_1 + n_i T_{i2} \underline{e}_2 + n_i T_{i3} \underline{e}_3$$

$$= (n_1 T_{11} + n_2 T_{21} + n_3 T_{31}) \underline{e}_1$$

$$+ (n_1 T_{12} + n_2 T_{22} + n_3 T_{32}) \underline{e}_2$$

$$+ (n_1 T_{13} + n_2 T_{23} + n_3 T_{33}) \underline{e}_3$$

Tensor : Tensor

A and B are tensors, What is A : B?

$$\underline{\underline{A}} : \underline{\underline{B}} = A_{ij} \underline{e}_i \underline{e}_j : B_{kl} \underline{e}_k \underline{e}_l$$

Nesting convention

$$= A_{ij} \underline{e}_i \cdot B_{kl} \delta_{jk} \underline{e}_l$$

$$= A_{ij} B_{kl} \delta_{jk} \delta_{il}$$

$$= A_{ij} B_{ji} \text{ or } A_{lk} B_{kl} \quad (\text{Scalar})$$

Sum over $i=1, 2$ and 3 ; $j=1, 2$, and 3

(9 Components)

Tensor : Tensor, cont'd

A and B are Tensors, What is A : B

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \bullet \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix} = A_{ij} B_{ji}$$

$$\begin{aligned} &= A_{11} B_{11} + A_{12} B_{21} + A_{13} B_{31} \\ &\quad + A_{21} B_{12} + A_{22} B_{22} + A_{23} B_{32} \\ &\quad + A_{31} B_{13} + A_{32} B_{23} + A_{33} B_{33} \end{aligned}$$

Invariants of Tensor $\underline{\underline{T}}$

First Invariant

$$\begin{aligned} I_T &= \text{trace}(\underline{\underline{T}}) \\ &= T_{11} + T_{22} + T_{33} \end{aligned}$$

Second Invariant

$$II_T = \frac{1}{2} \left[(I_T)^2 - \text{trace}(\underline{\underline{T}} \cdot \underline{\underline{T}}) \right]$$

Third Invariant

$$\begin{aligned} III_T &= \det(\underline{\underline{T}}) = \det \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \\ &= T_{11} (T_{22} T_{33} - T_{23} T_{32}) \\ &\quad - T_{12} (T_{21} T_{33} - T_{31} T_{23}) + T_{13} (T_{21} T_{32} - T_{31} T_{22}) \end{aligned}$$

All invariants –scalar & independent of coordinate system

Gradient ($\underline{\nabla}$)

$$\begin{aligned}\underline{\nabla} &\equiv \frac{\partial}{\partial x_i} \underline{e}_i \\ &= \frac{\partial}{\partial x_1} \underline{e}_1 + \frac{\partial}{\partial x_2} \underline{e}_2 + \frac{\partial}{\partial x_3} \underline{e}_3\end{aligned}$$

For Example gradient of density ρ

$$\begin{aligned}\underline{\nabla} \rho &\equiv \frac{\partial \rho}{\partial x_i} \underline{e}_i \\ &= \frac{\partial \rho}{\partial x_1} \underline{e}_1 + \frac{\partial \rho}{\partial x_2} \underline{e}_2 + \frac{\partial \rho}{\partial x_3} \underline{e}_3\end{aligned}$$

Operational Order !

Operation	Order
Vector	1
Tensor	2
Scalar	0
Dot Product	-2
Double Dot Product	-4
Cross Product	-1

Examples:

$$\underline{\underline{A}} \cdot \underline{\underline{A}}$$

Order =

$$\underline{\underline{A}} : \underline{\underline{A}}$$

Order =

$$\underline{V} \cdot \underline{A}$$

Order =

$$\underline{V} \underline{W}$$

Order =

Operational Order !

Operation	Order
Vector	1
Tensor	2
Scalar	0
Dot Product	-2
Double Dot Product	-4
Cross Product	-1

Examples:

$$\underline{\underline{A}} \cdot \underline{\underline{A}}$$

$$\text{Order} = 2+2-2=2 \quad (\text{Tensor})$$

$$\underline{\underline{A}}:\underline{\underline{A}}$$

$$\text{Order} = 2+2-4=0 \quad (\text{scalar})$$

$$\underline{V} \cdot \underline{A}$$

$$\text{Order} = 1+2-2=1 \quad (\text{Vector})$$

$$\underline{V} \underline{W}$$

$$\text{Order} = 1+1=2 \quad (\text{Tensor})$$

$$= V_i W_j \underline{e}_i \underline{e}_j$$

So far.....

- **Concepts of Vectors & Tensors**

Index notation

Dot and double dot products

Orders of vector, tensor & various operation

- **Properties of Stress Tensor T**

Symmetric

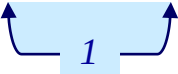
Normal stress – diagonal elements

Shear stress – off-diagonal elements

Scalar invariants of tensor

Class Exercise

$\underline{\underline{A}} \bullet \underline{\underline{B}} = ?$ (write simplified version in index notation)

$$\underline{\underline{A}} \bullet \underline{\underline{B}} = A_{ij} \underline{e}_i \underline{e}_j \bullet B_{kl} \underline{e}_k \underline{e}_l$$


$$= A_{ij} \underline{e}_i B_{kl} \delta_{jk} \underline{e}_l$$

$$= A_{ij} B_{jl} \underline{e}_i \underline{e}_l$$

Now this is a tensor of the form

$$= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

What is term 2?

Class Exercise

$$\underline{\underline{A}} \bullet \underline{\underline{B}} = A_{ij} B_{jl} \underline{e}_i \underline{e}_l$$

Now this is a tensor of the form

this is the term with $\underline{e}_1 \underline{e}_2$ in it

$$= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \text{ What is term 2?}$$

$$= A_{1j} B_{j2} \underline{e}_1 \underline{e}_2$$

$$= [A_{11} B_{12} + A_{12} B_{22} + A_{13} B_{32}] \underline{e}_1 \underline{e}_2$$

Stress Tensor

Consider sample

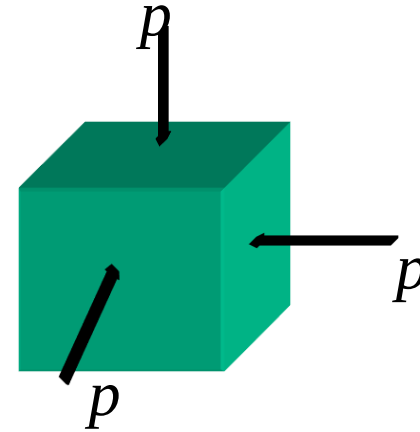
- under pressure or at ambient condition
- no flow



Can support only normal stresses

$$\underline{\underline{T}} = \begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix}$$

Static condition



$$\underline{\underline{T}} = -p\underline{\underline{I}}, \quad \underline{\underline{I}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Identity or Unit tensor

Stress tensor & flow

$$\underline{\underline{T}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

Total stress tensor *Viscous stress tensor*

$$\underline{\underline{\tau}} \rightarrow 0 \text{ (when no flow)}$$

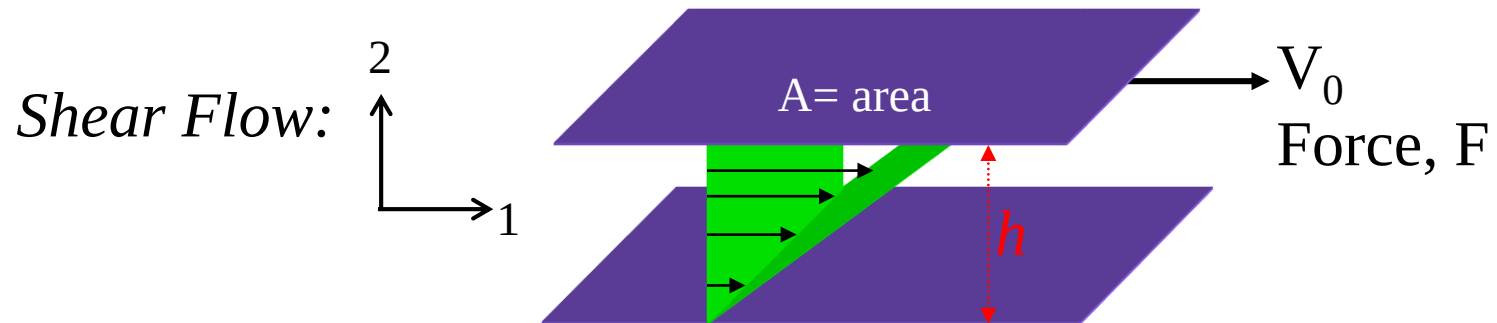
Advantage –isolate isotropic or hydrostatic pressure effects

For normal stress measurements, want to subtract “p” effect

Use normal stress difference, not normal stress:

$$T_{11} - T_{22} = \tau_{11} - \tau_{22}$$

Deformation description



$$\text{Stress} = \frac{F}{A} = \tau_{21}$$

$$\tau_{21} = \eta \left(\frac{v_o}{h} \right)$$

η = viscosity

$$\left(\frac{v_o}{h} \right) = \text{shear rate} = \dot{\gamma}$$

How do we get shear rate? Form of shear rate?

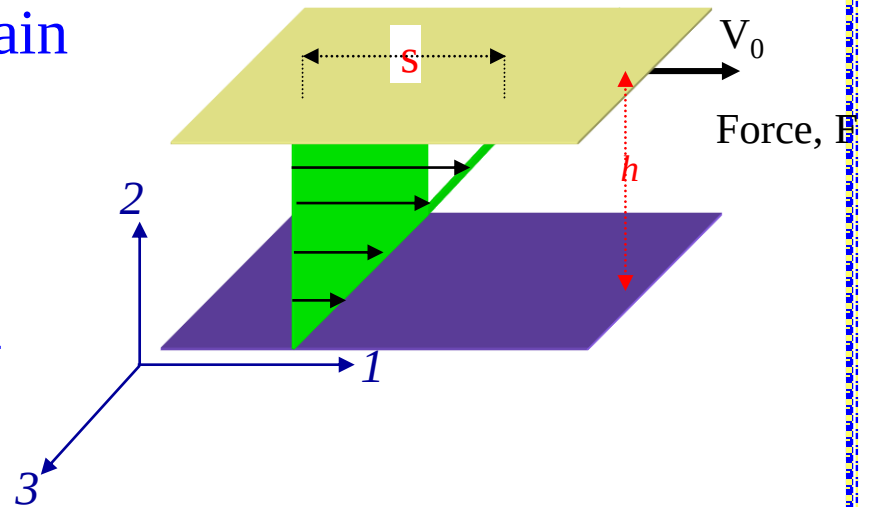
Deformation description (kinematics)

Stress = viscosity * rate of change of strain

$\left(\frac{ds}{dt}\right) = \text{Rate of change of strain}$

$$\frac{ds}{dt} = v_o$$

$$\text{Rate of change of strain} = \frac{v_o}{h}$$



Deformation description (kinematics)

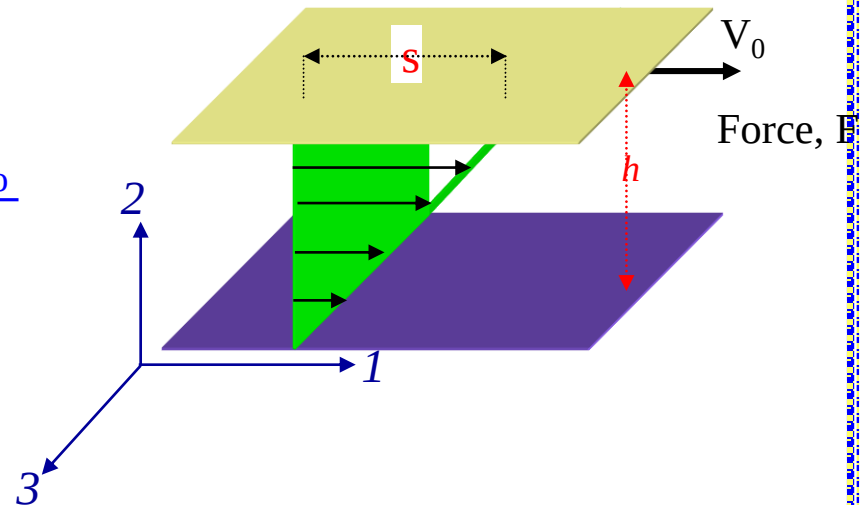
$$\text{Rate of change of strain} = \frac{V_o}{h}$$

In general:

$$\tau_{12} = \eta \frac{\partial v_1}{\partial x_2}$$

in TENSORIAL form

$$\underline{\underline{\tau}} = \eta \underline{\underline{\dot{\gamma}}} = \eta 2 \underline{\underline{D}}$$



Stress Tensor, cont'd

$$\underline{\underline{\tau}} = \eta \underline{\underline{\dot{\gamma}}} = \eta 2 \underline{\underline{D}}$$

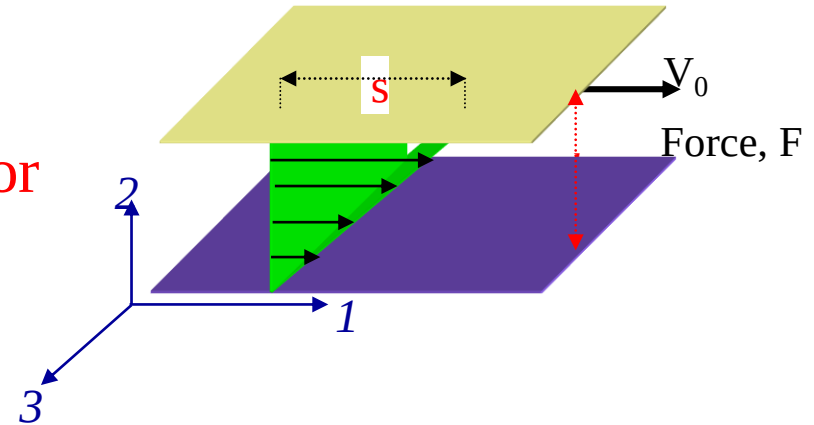
$$2\underline{\underline{D}} \equiv \underline{\underline{\dot{\gamma}}} \equiv \text{Rate of strain tensor}$$

$$\dot{\gamma}_{ij} = \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right)$$

$$\underline{v} = (v_1, 0, 0)$$

$$v_1 = \frac{V_0}{h} x_2$$

$$\begin{bmatrix} 0 & \tau_{12} & 0 \\ \tau_{21} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \eta \begin{bmatrix} 0 & \frac{\partial v_1}{\partial x_2} & 0 \\ \frac{\partial v_1}{\partial x_2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Rate of Strain Tensor

$$\dot{\gamma}_{ij} = \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right)$$

$$v_1 = \frac{V_0}{h} x_2$$

$$\dot{\gamma}_{11} = ?$$

$$\dot{\gamma}_{32} = ?$$

$$\dot{\gamma}_{12} = ?$$

$$\dot{\gamma}_{21} = ?$$

Find the different components of the rate of strain tensor.

$$\dot{\gamma}_{21} = ?$$

Rate of Strain Tensor

$$\dot{\gamma}_{ij} = \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right)$$

$$v_1 = \frac{V_0}{h} x_2$$

$$\dot{\gamma}_{11} = \frac{\partial v_1}{\partial x_1} + \frac{\partial v_1}{\partial x_1} = 2 \frac{\partial v_1}{\partial x_1} = 0$$

$$\dot{\gamma}_{32} = \frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_2} = 0$$

$$\dot{\gamma}_{12} = \cancel{\frac{\partial v_2}{\partial x_1}} + \frac{\partial v_1}{\partial x_2} = \frac{\partial v_1}{\partial x_2}$$

$$\dot{\gamma}_{21} = \frac{\partial v_1}{\partial x_2} + \cancel{\frac{\partial v_2}{\partial x_1}} = \frac{\partial v_1}{\partial x_2}$$

*Shear rate not a function of position –
homogeneous flow*

$$\dot{\gamma}_{21} = \dot{\gamma}_{12} = \frac{\partial v_1}{\partial x_2} = \frac{V_0}{h} = \dot{\gamma}$$

Transition to new topic

- End of Quick Overview
- Start on Chap 3 of Dynamics of Polymeric Liquids
- Flow types & Material Functions

Material Functions

Inherent properties of materials

Serves to classify material

Used in developing constitution equations

Provides information – microstructural or molecular level

Material Functions, cont'd

Newtonian Fluids

(oil, water,)

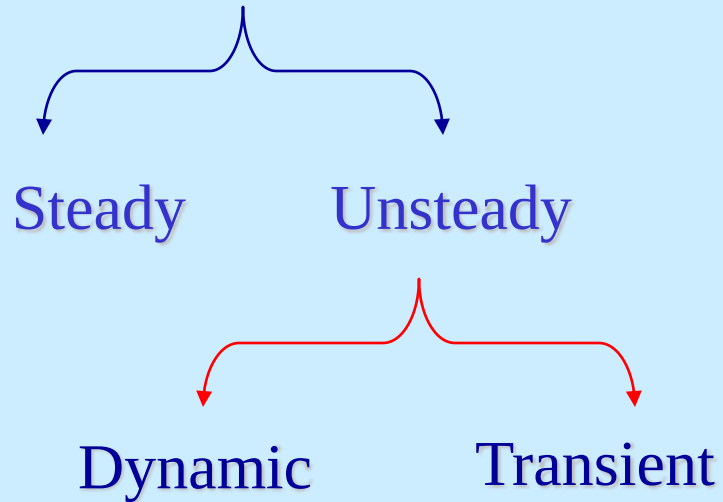
- ρ, η (constant)
 - Independent of flow type and shear rate
- η characterizes Newtonian fluids

Complex Fluids

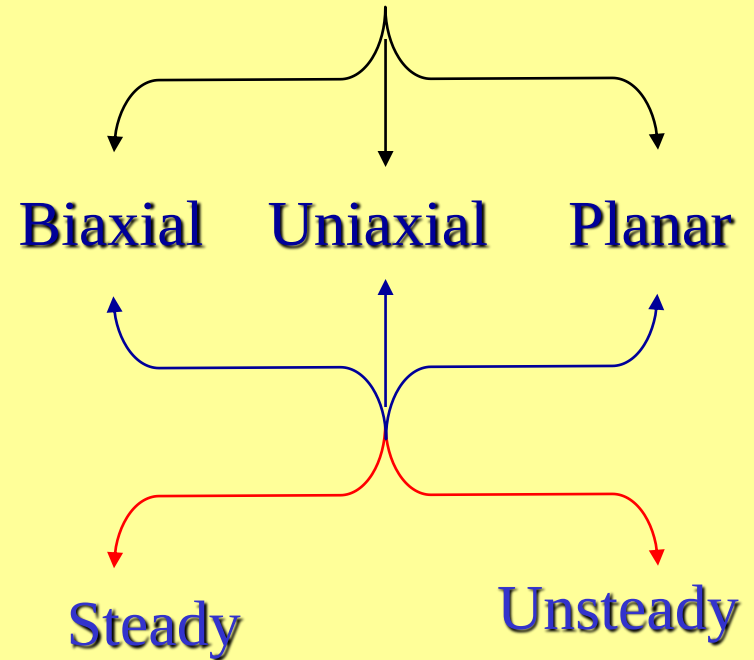
- Need many material properties to be fully characterized
- Material properties are functions of flow type or flow rate

Flow Type

Shear Flow

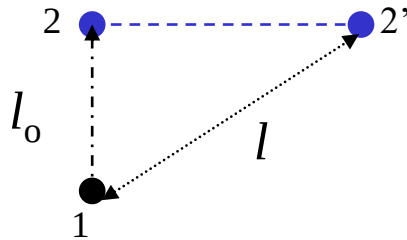


Extensional Flow



Features of Shear Flow

- Family of surfaces that slide past each other

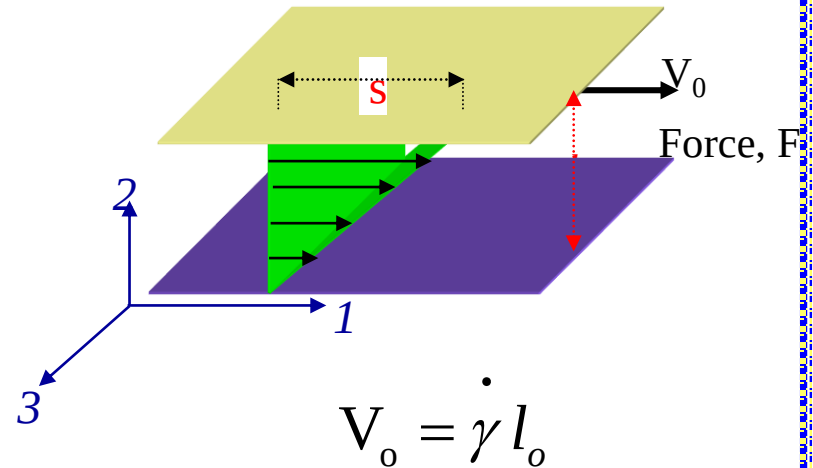


$$l = \sqrt{l_o^2 + l_o^2 \dot{\gamma}^2 t^2}$$

$$\cong l_o \dot{\gamma} t$$

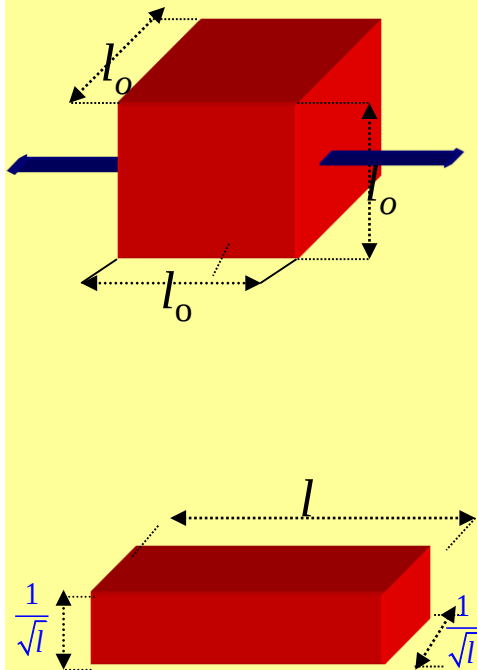
(for larger times)

- Distance between neighboring fluid particles change linearly with time

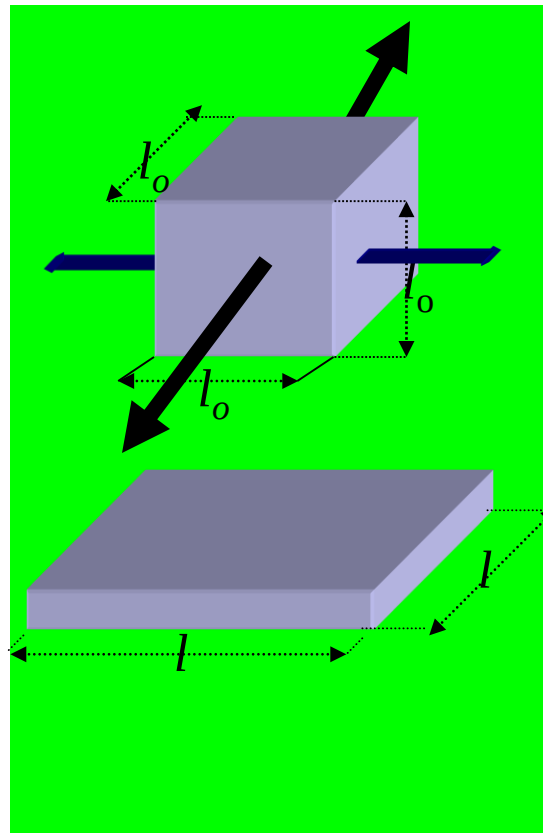


Extensional Flow

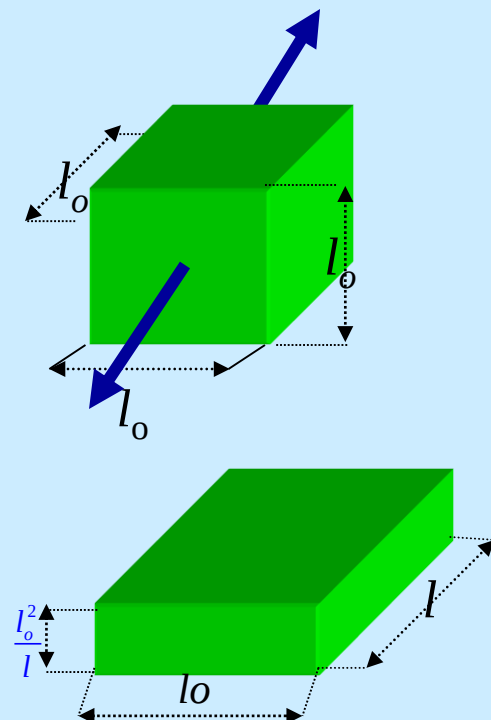
Uniaxial



Biaxial



Planar



Extensional Flow

$$v_x = \dot{\epsilon}x, \quad \dot{\epsilon} \text{ is the extension rate}$$

$$v_x = \dot{\epsilon}l$$

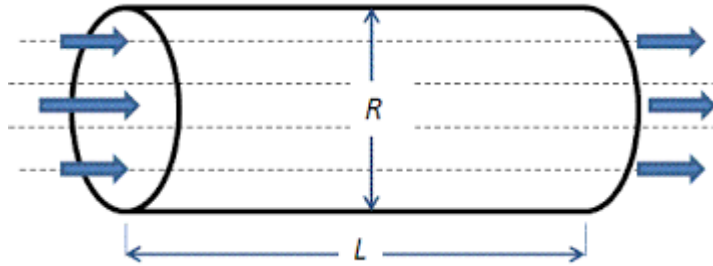
$$\frac{dl}{dt} = \dot{\epsilon}l$$

$$\ln l = \dot{\epsilon}t, \quad t = 0 \quad l = l_o$$

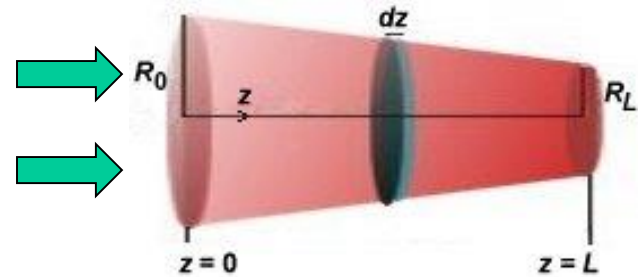
$$l = l_o \exp(\dot{\epsilon}t)$$

- Neighboring particles move relative to each other at an exponential rate → “**strong**” flow
- Extensional Flows are volume preserving

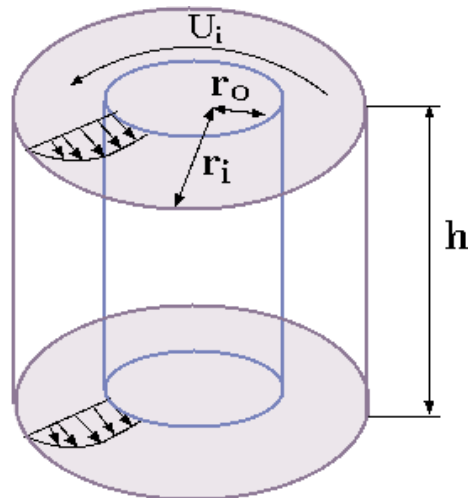
What kind of flow are these?



Flow through pipe



Flow through tapered pipe



Flow between concentric cylinders

END OF CLASS

Stress tensors in different flows

Extension :

$$\underline{\underline{T}} = \begin{bmatrix} -p + \tau_{xx} & 0 & 0 \\ 0 & -p + \tau_{yy} & 0 \\ 0 & 0 & -p + \tau_{zz} \end{bmatrix}$$

Shear:

$$\underline{\underline{T}} = \begin{bmatrix} -p + \tau_{xx} & \tau_{yx} & 0 \\ \tau_{xy} & -p + \tau_{yy} & 0 \\ 0 & 0 & -p + \tau_{zz} \end{bmatrix}$$

$$\tau_{xz} = \tau_{yz} = 0$$

i.e., only one shear component non-zero

For either flow,
normal stresses of interest

$$T_{xx} - T_{yy} = \tau_{xx} - \tau_{yy}$$

$$T_{yy} - T_{zz} = \tau_{yy} - \tau_{zz}$$

Material function- steady shear flow

3 material functions characterize a material

$$\eta(\dot{\gamma}) = \frac{\tau_{yx}}{\dot{\gamma}}$$

Viscosity

$$\psi_1(\dot{\gamma}) = \frac{\tau_{xx} - \tau_{yy}}{\dot{\gamma}^2} = \frac{N_1}{\dot{\gamma}^2}$$

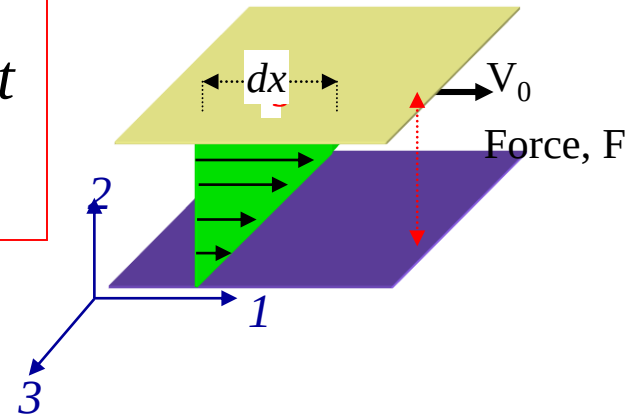
First Normal Stress Coefficient

$$\psi_2(\dot{\gamma}) = \frac{\tau_{yy} - \tau_{zz}}{\dot{\gamma}^2} = \frac{N_2}{\dot{\gamma}^2}$$

Second Normal Stress Coefficient

Steady Shear Flow

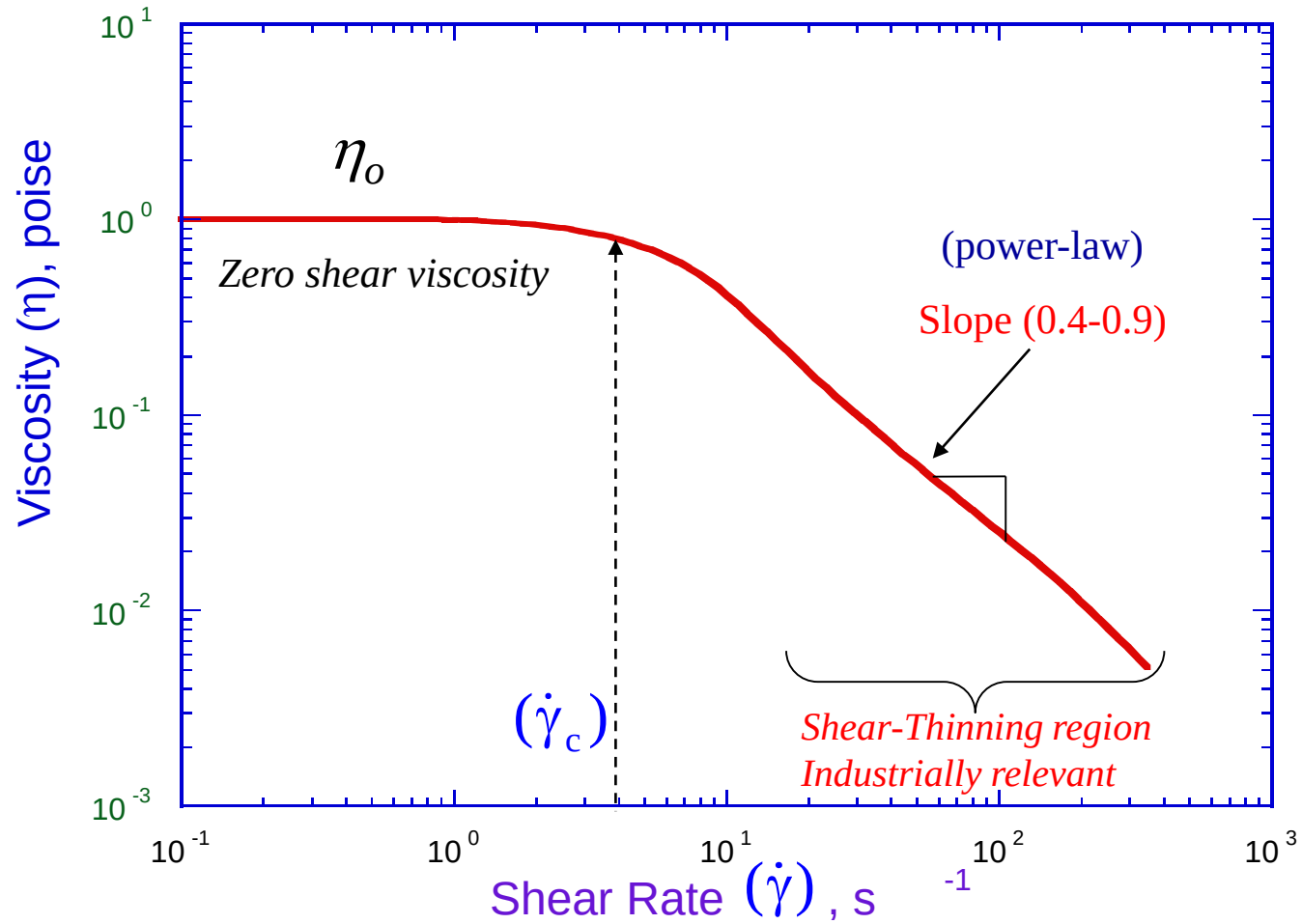
- $x(1)$ Flow Direction
- $y(2)$ Direction of Velocity Gradient
- $z(3)$ Neutral Direction



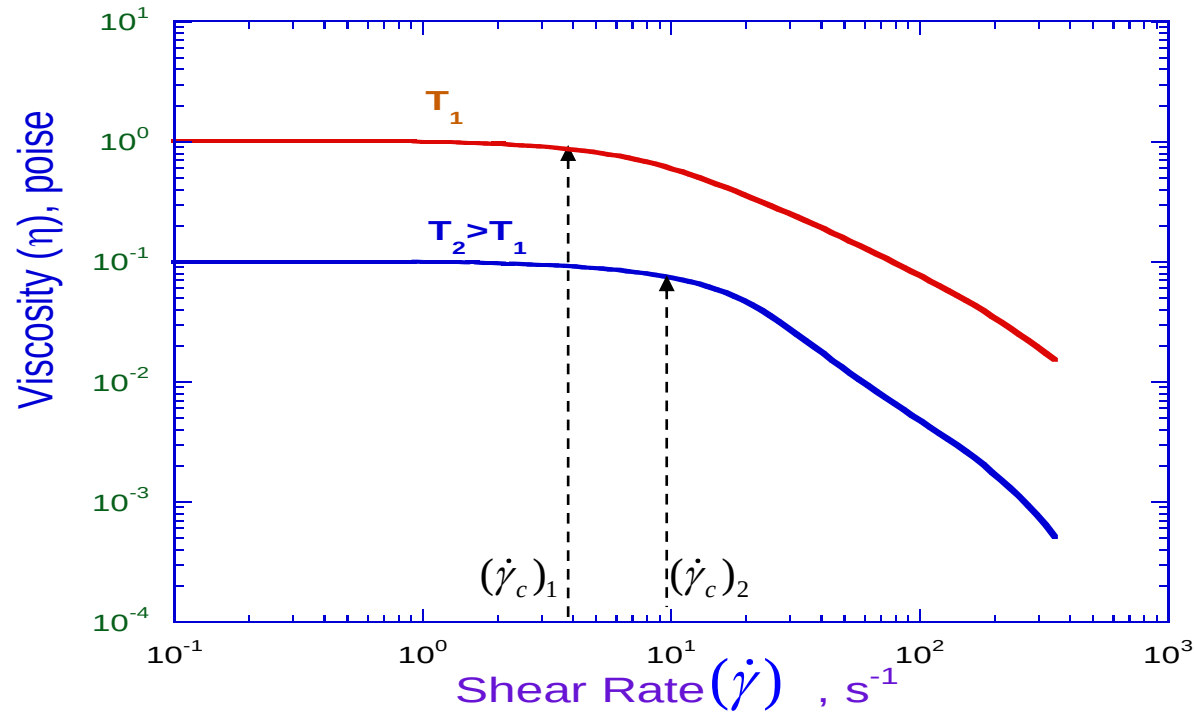
3 Stresses of Interest in Steady Shear Flow

- τ_{yx} Shear Stress
- $N_1 = \tau_{xx} - \tau_{yy}$ First Normal Stress Difference
- $N_2 = \tau_{yy} - \tau_{zz}$ Second Normal Stress Difference

Shear Viscosity



Temperature effect

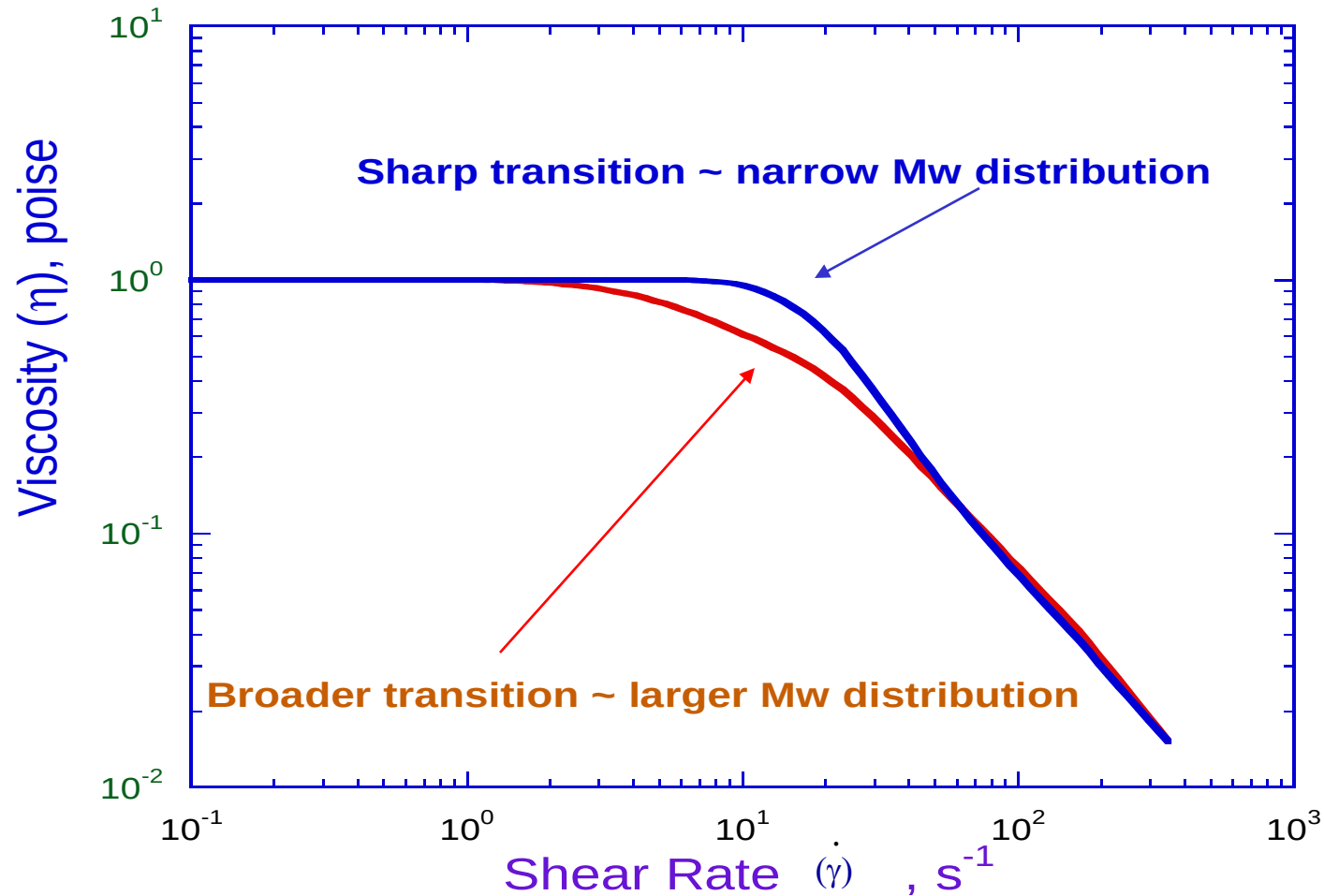


As Temperature $\uparrow$

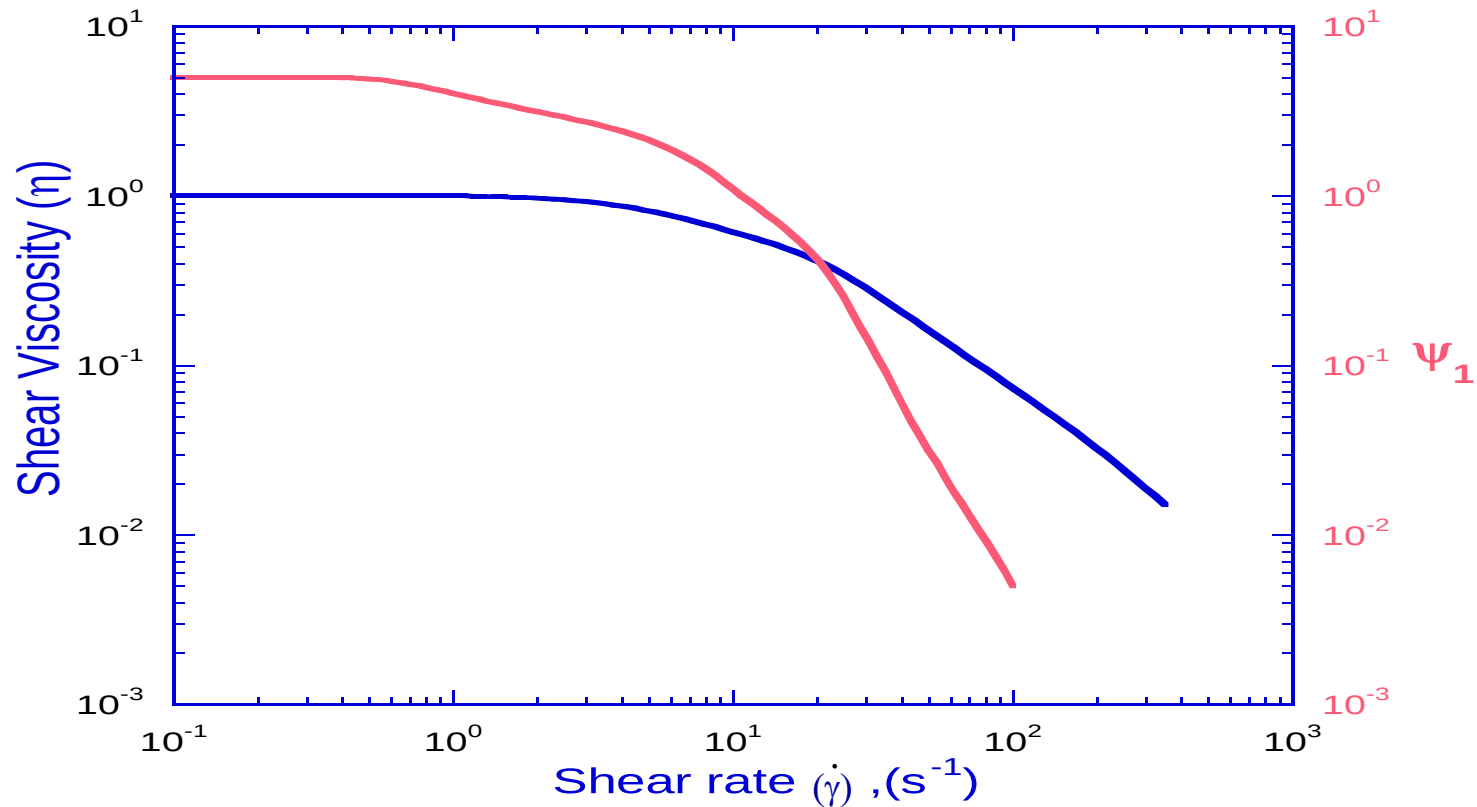
$\eta_0 \downarrow$ and $(\dot{\gamma}_c) \uparrow$

$\lambda \downarrow$ (Longest relaxation time of material $\cong 1/(\dot{\gamma}_c)$)

Molecular Weight Distribution



η versus Normal Stress Coefficient



Slope of normal Stress Coefficient is steeper than that of viscosity